

Toronto Rush Hour Traffic Analysis*

INF2167 Project - Group 7

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abstract goes here

*Code and data are available at: https://github.com/srg-uoft/INF2167_Group7_Project.

1 Introduction

Welcome...

2 Research Questions(s) & Hypothesis

This project will aim to address some research questions and preeminent assumptions about traffic patterns, as discussed below.

2.1 Questions

Our research questions are...

2.2 Hypotheses

Some hypotheses we have are... These are of interest because...

3 Relevant Literature

According to (Alexander 2023)...

congestion in Canadian cities not only costs time and sacrafices productivities, but also increases environmental burdens by more greenhouse gas emission. (<https://comt.ca/Reports/High%20Cost%20of%20Congestion%202012.pdf>)

4 Dataset & Methods

4.1 Dataset

4.1.1 Source

Dataset obtained from (City of Toronto, n.d.).

4.1.2 Structure

This is how our dataset is structured

4.1.3 Key Variables

Key variables from our dataset are...

4.2 Methods

Methods we plan to apply in our analysis are...

5 Exploratory Data Analysis

5.1 Overview

Initial exploratory data analysis was done using statistical programming language (R Core Team 2023) and the Tidyverse library (Wickham et al. 2019), with the goal of uncovering high-level trends and possible variables of interest for further analysis. Key discoveries which we hope will aid in answering our research question are elaborated on below.

5.2 Key Aspect 1

One key dataset aspect that relates to our variables of interest is...

5.3 Key Aspect 2

Another key dataset aspect that relates to our variables of interest is...

5.4 Key Aspect 3

Another key dataset aspect that relates to our variables of interest is...

6 Next Steps

Having outlined the research question of interest, target data set, and preliminary exploratory data analysis above, we will move into more detailed and thorough analysis of the dataset to identify any trends which could help address the primary research question. Conclusions and discussion, along with raw results and any models used in their generation, will be presented in a final report and oral presentation in August 2026.

7 References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
- City of Toronto. n.d. *Traffic Volumes - Midblock Vehicle Speed, Volume and Classification Counts*. Accessed May 30, 2026. <https://open.toronto.ca/dataset/traffic-volumes-midblock-vehicle-speed-volume-and-classification-counts/>.
- R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.